

H0 — decision summary (1/2)

llama31-8b (ctx 131,072, 1,024 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 4.046 b cont 4.063 b paired delta -0.010 b (need >= 0.1)
heads with niah > cont 39.9% (need >= 60%, n=1,024)
HEADS IN BAND 69.6% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 2.50x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.2x p90 4.7x
ladder width median 4.06 b IQR [3.39, 4.71] 100% of heads above 1.4
tau median 2.85 excl. sinks 2.85
top-1 weight median 0.0165 95% mass in 24924 tokens (20.7% of ctx)
B=2b vs uniform 16.0x vs eviction 2.7x vs BEST corner 2.6x evicts 62%
B=3b vs uniform 15.8x vs eviction 3.2x vs BEST corner 2.8x evicts 49%
quantizer alpha @2b 0.941 (1.0 = no temperature distortion); spearman@2b 0.607 [H2 answered]
heads where eviction beats a 1-bit tier: 94%
heads where eviction beats a 2-bit tier: 13%
linearization check (predicted/measured gain @3b): 1.82 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta -0.010 b < 0.1 b; only 39.9% of heads have niah > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

llama33-70b (ctx 131,072, 5,120 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 1.727 b cont 1.729 b paired delta -0.002 b (need >= 0.1)
heads with niah > cont 43.3% (need >= 60%, n=5,120)
HEADS IN BAND 39.8% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.98x (geometric mean of max(gain,1) over all heads)
in-band heads median 4.1x p90 10.9x
ladder width median 1.73 b IQR [1.30, 2.17] 70% of heads above 1.4
tau median 1.23 excl. sinks 1.23
top-1 weight median 0.0008 95% mass in 74701 tokens (61.9% of ctx)
B=2b vs uniform 2.3x vs eviction 7.2x vs BEST corner 2.1x evicts 33%
B=3b vs uniform 1.6x vs eviction 9.7x vs BEST corner 1.5x evicts 16%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.633 [H2 answered]
heads where eviction beats a 1-bit tier: 13%
heads where eviction beats a 2-bit tier: 4%
linearization check (predicted/measured gain @3b): 3.13 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta -0.002 b < 0.1 b; only 43.3% of heads have niah > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

mistral-7b (ctx 32,768, 1,024 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 2.674 b cont 2.665 b paired delta +0.002 b (need >= 0.1)
heads with niah > cont 53.8% (need >= 60%, n=1,024)
HEADS IN BAND 84.5% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 4.06x (geometric mean of max(gain,1) over all heads)
in-band heads median 4.6x p90 10.5x
ladder width median 2.68 b IQR [2.32, 3.12] 99% of heads above 1.4
tau median 1.94 excl. sinks 1.94
top-1 weight median 0.0053 95% mass in 10432 tokens (34.6% of ctx)
B=2b vs uniform 6.2x vs eviction 3.7x vs BEST corner 3.1x evicts 53%
B=3b vs uniform 6.9x vs eviction 6.6x vs BEST corner 4.2x evicts 36%
quantizer alpha @2b 0.940 (1.0 = no temperature distortion); spearman@2b 0.824 [H2 answered]
heads where eviction beats a 1-bit tier: 67%
heads where eviction beats a 2-bit tier: 3%
linearization check (predicted/measured gain @3b): 2.34 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta +0.002 b < 0.1 b; only 53.8% of heads have niah > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

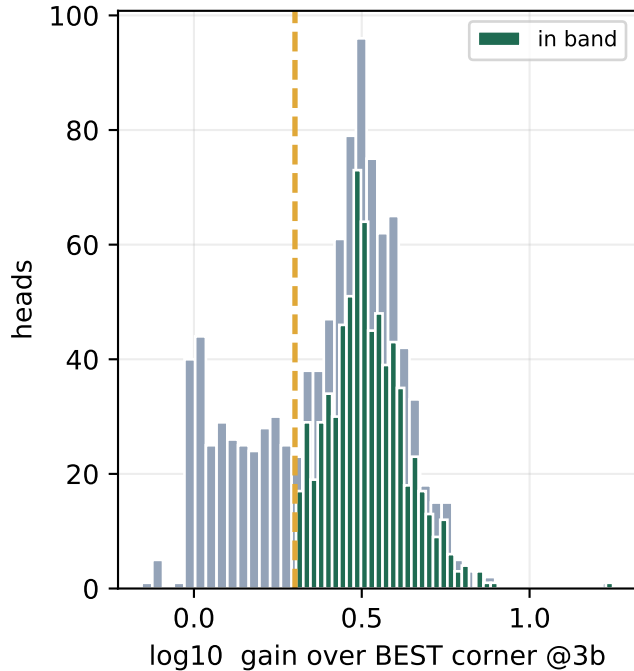
H0 — decision summary (2/2)

qwen15-moe-a2.7b (ctx 32,768, 384 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 4.291 b cont 4.321 b paired delta +0.028 b (need >= 0.1)
heads with niah > cont 69.8% (need >= 60%, n=384)
HEADS IN BAND 68.8% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 2.58x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.4x p90 5.3x
ladder width median 4.29 b IQR [3.54, 5.33] 99% of heads above 1.4
tau median 3.04 excl. sinks 3.04
top-1 weight median 0.0173 95% mass in 7971 tokens (26.4% of ctx)
B=2b vs uniform 7.7x vs eviction 3.0x vs BEST corner 2.6x evicts 61%
B=3b vs uniform 9.7x vs eviction 3.9x vs BEST corner 2.9x evicts 47%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.582 [H2 answered]
heads where eviction beats a 1-bit tier: 84%
heads where eviction beats a 2-bit tier: 7%
linearization check (predicted/measured gain @3b): 2.42 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta +0.028 b < 0.1 b). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

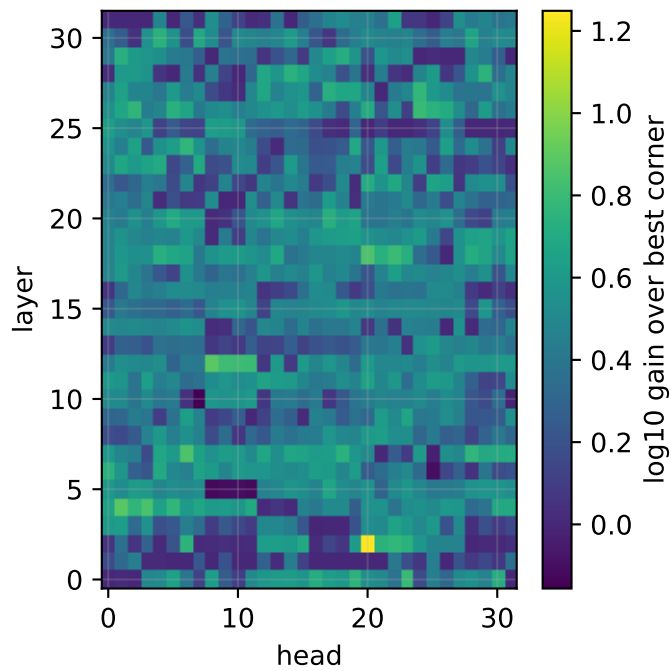
qwen3-30b-a3b-2507 (ctx 131,072, 1,536 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 4.602 b cont 4.498 b paired delta +0.071 b (need >= 0.1)
heads with niah > cont 64.0% (need >= 60%, n=1,536)
HEADS IN BAND 14.2% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.27x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.4x p90 3.1x
ladder width median 4.55 b IQR [3.93, 5.46] 100% of heads above 1.4
tau median 3.15 excl. sinks 3.15
top-1 weight median 0.1189 95% mass in 2550 tokens (2.1% of ctx)
B=2b vs uniform 59.5x vs eviction 1.3x vs BEST corner 1.3x evicts 68%
B=3b vs uniform 33.6x vs eviction 1.1x vs BEST corner 1.1x evicts 54%
quantizer alpha @2b 0.941 (1.0 = no temperature distortion); spearman@2b 0.492 [H2 answered]
heads where eviction beats a 1-bit tier: 100%
heads where eviction beats a 2-bit tier: 75%
linearization check (predicted/measured gain @3b): 0.87 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta +0.071 b < 0.1 b). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

qwen3-8b (ctx 40,960, 1,152 heads)
INPUT VALIDITY FAIL haystack real
ladder niah 4.421 b cont 4.441 b paired delta -0.011 b (need >= 0.1)
heads with niah > cont 41.7% (need >= 60%, n=1,152)
HEADS IN BAND 54.0% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.94x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.9x p90 3.9x
ladder width median 4.42 b IQR [3.62, 5.92] 100% of heads above 1.4
tau median 3.15 excl. sinks 3.15
top-1 weight median 0.0150 95% mass in 6240 tokens (16.6% of ctx)
B=2b vs uniform 13.5x vs eviction 2.1x vs BEST corner 2.0x evicts 64%
B=3b vs uniform 15.1x vs eviction 2.5x vs BEST corner 2.2x evicts 50%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.466 [H2 answered]
heads where eviction beats a 1-bit tier: 99%
heads where eviction beats a 2-bit tier: 39%
linearization check (predicted/measured gain @3b): 1.76 (1.0 = exact)
VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta -0.011 b < 0.1 b; only 41.7% of heads have niah > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context behaviour. See [h0_measurement/bugs/1_from_synthetic_to_real_corpus/](#).

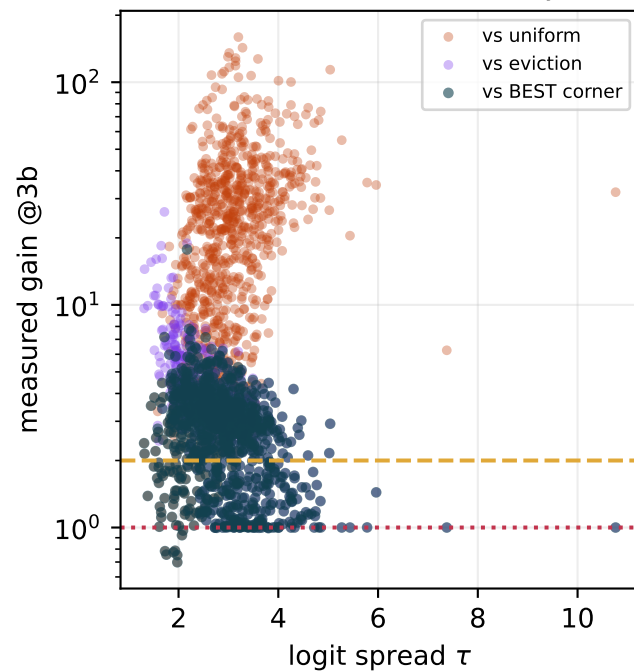
HEADS IN BAND: 70%



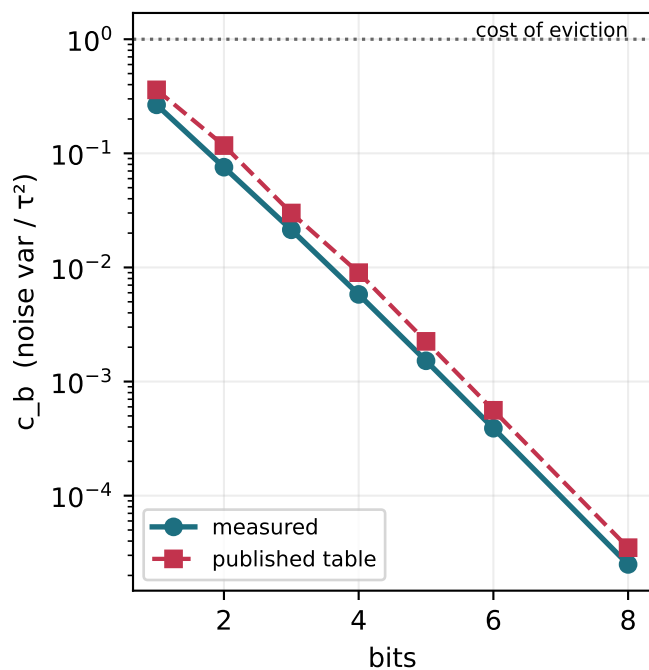
Which heads the router sends to SIEVE



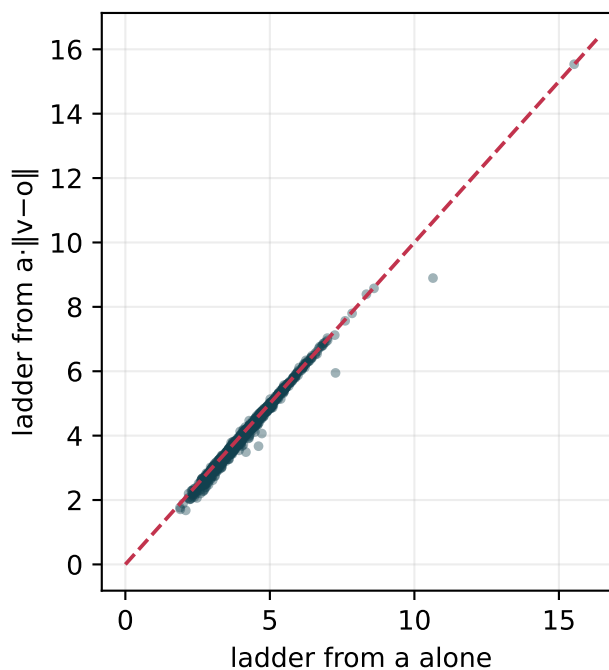
Where does the interior actually win?



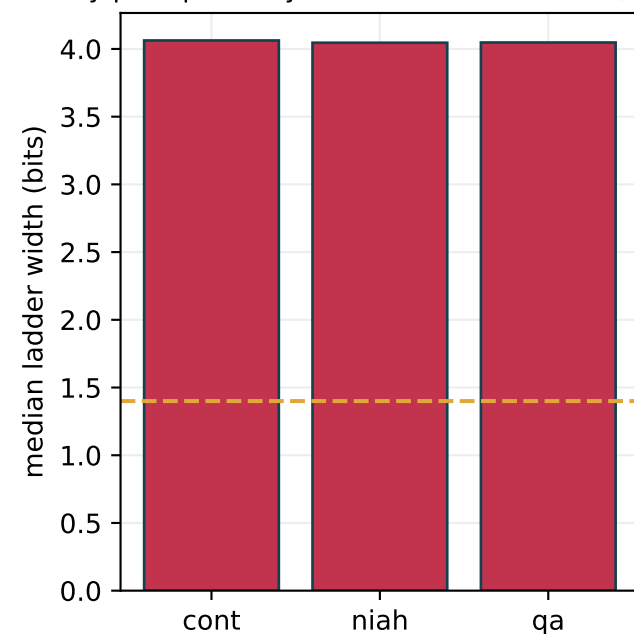
Measured vs assumed noise



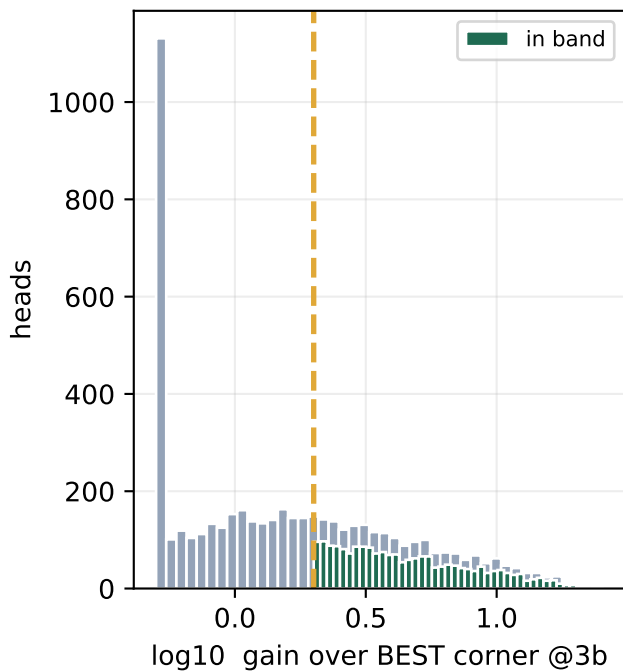
Does the value term matter?



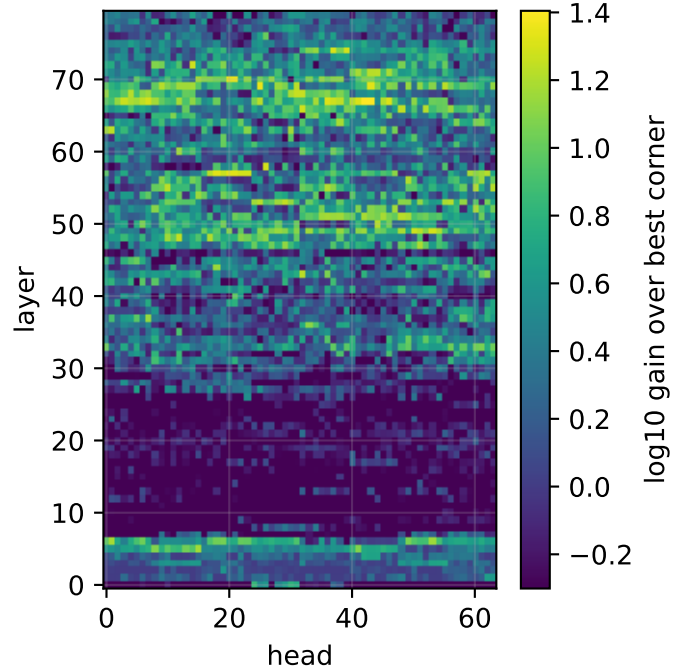
By prompt family — niah—cont -0.010 b, FA



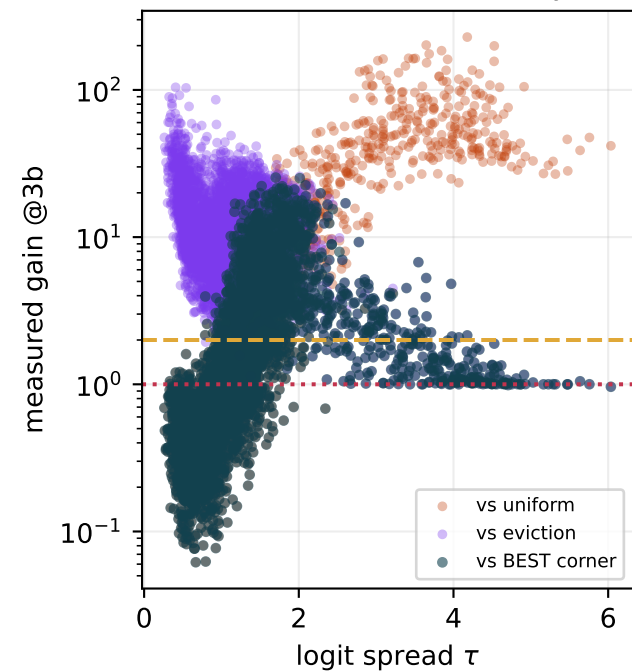
HEADS IN BAND: 40%



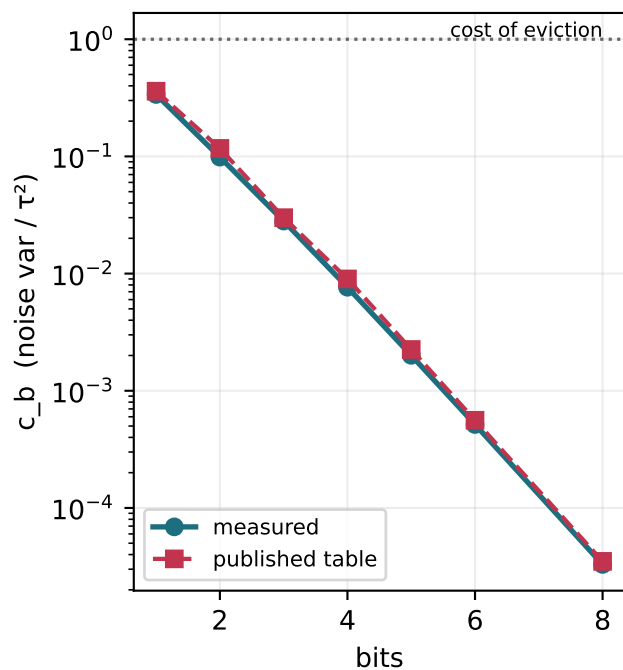
Which heads the router sends to SIEVE



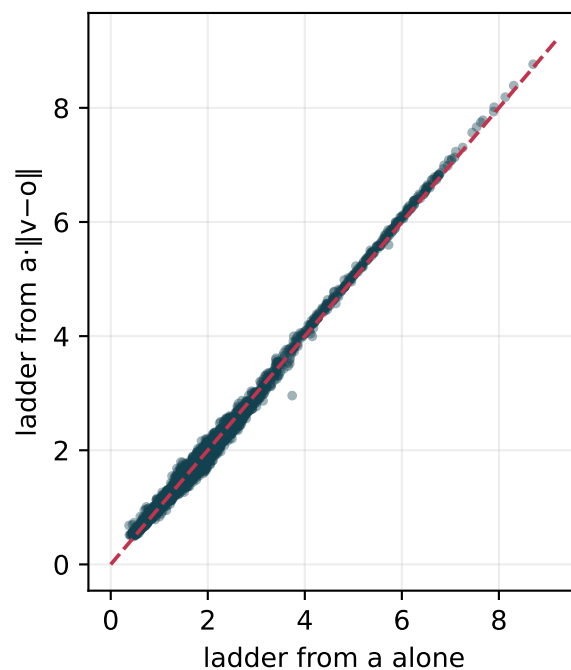
Where does the interior actually win?



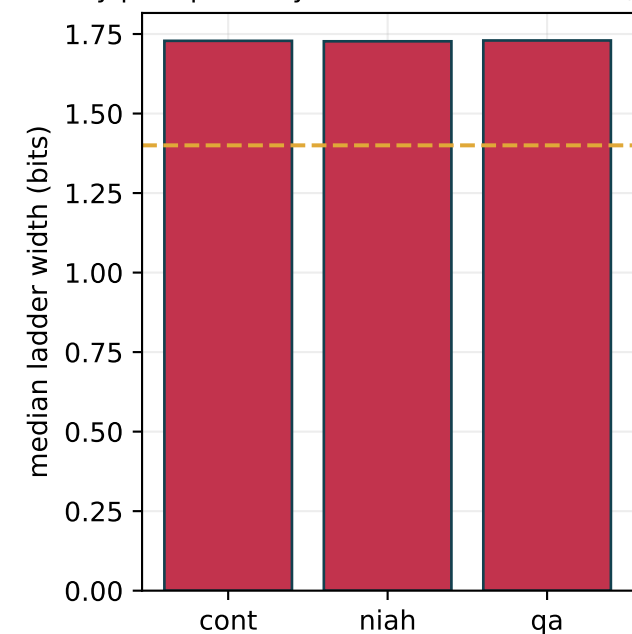
Measured vs assumed noise



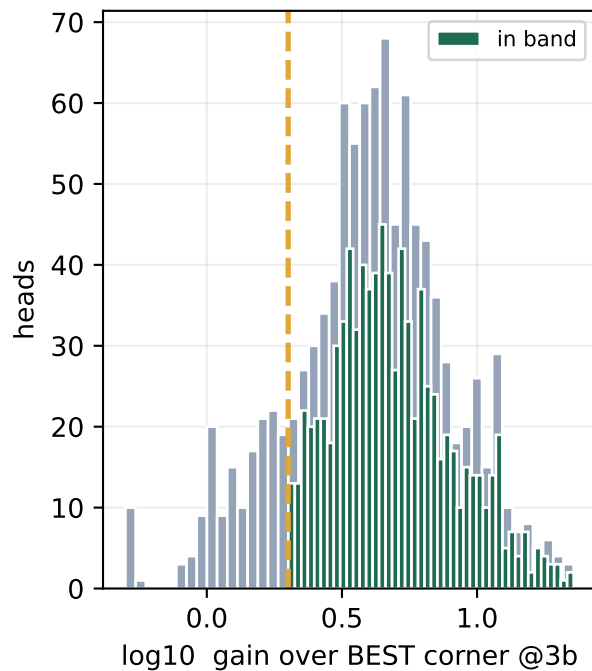
Does the value term matter?



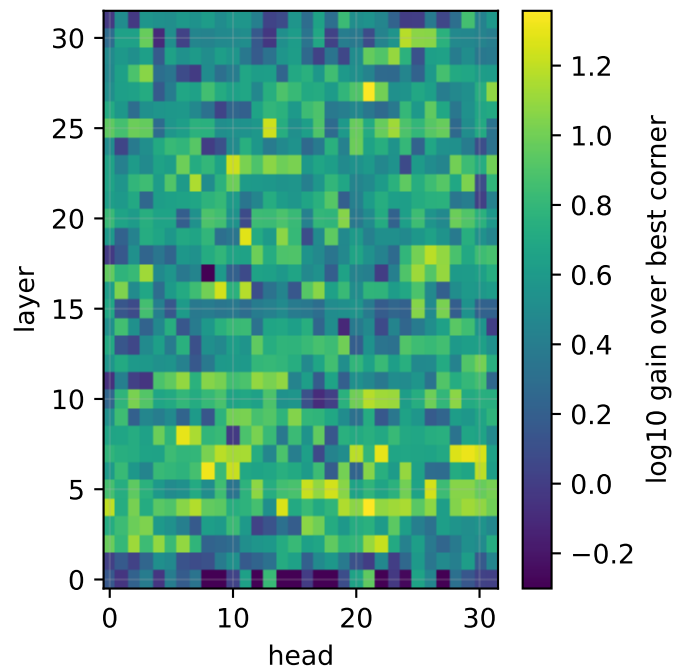
By prompt family — niah-cont -0.002 b, F



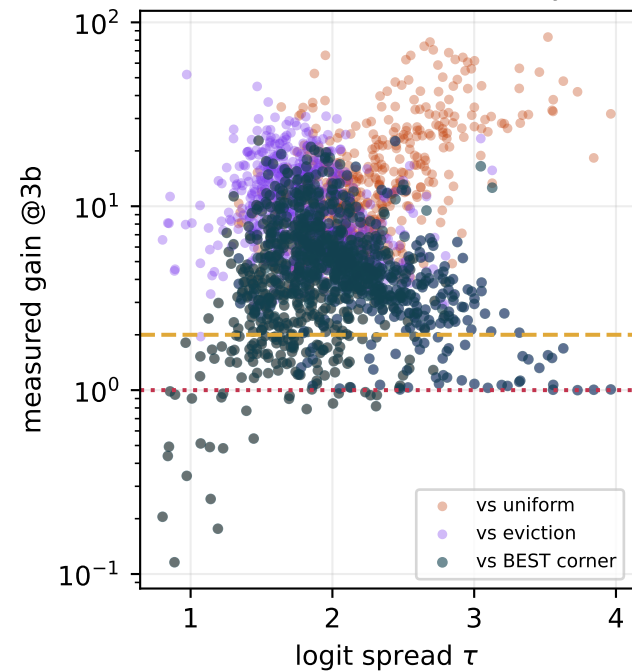
HEADS IN BAND: 84%



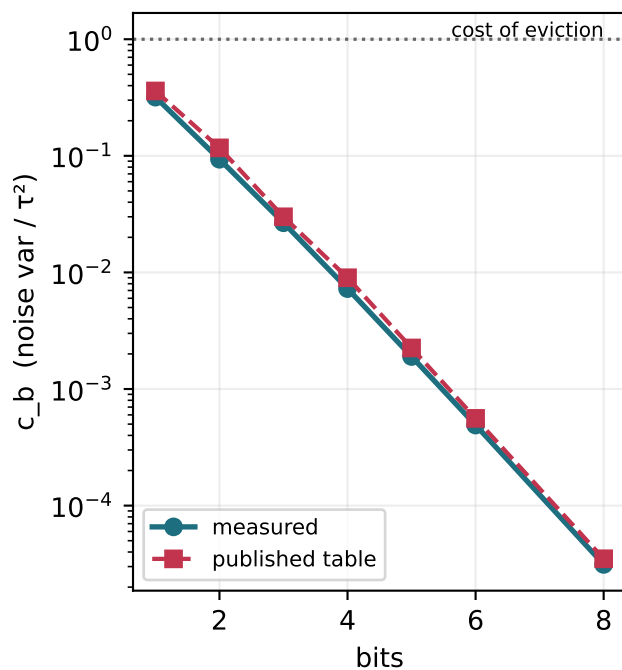
Which heads the router sends to SIEVE



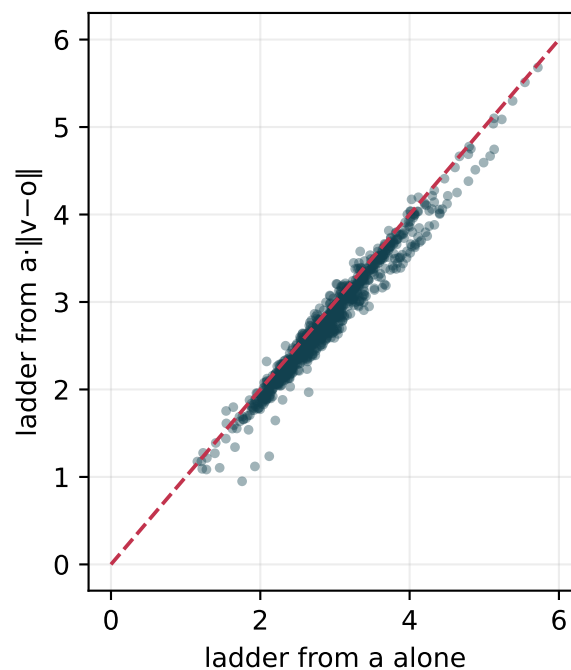
Where does the interior actually win?



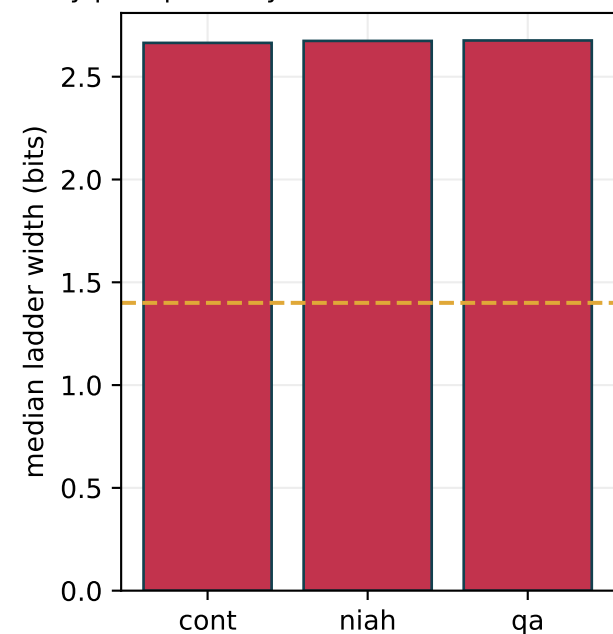
Measured vs assumed noise



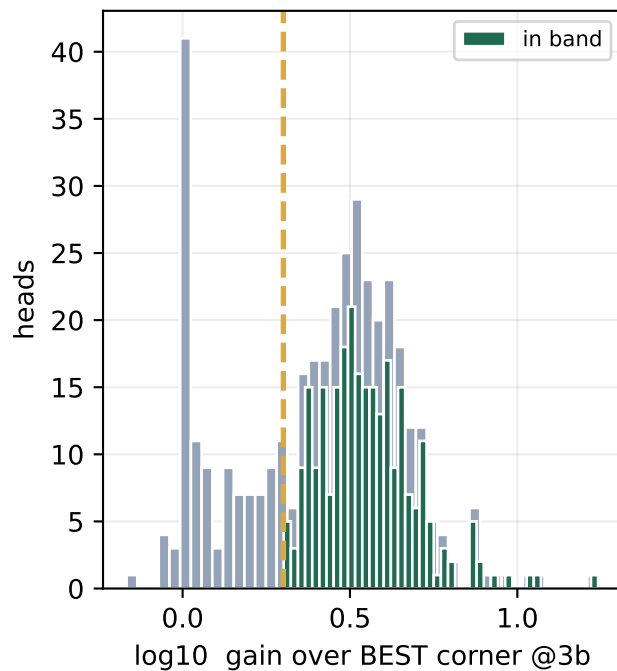
Does the value term matter?



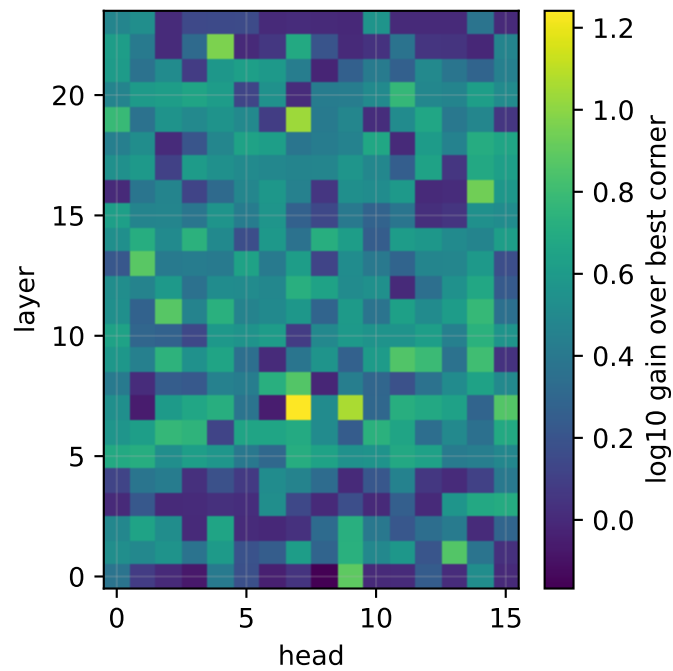
By prompt family — niah-cont +0.002 b, F



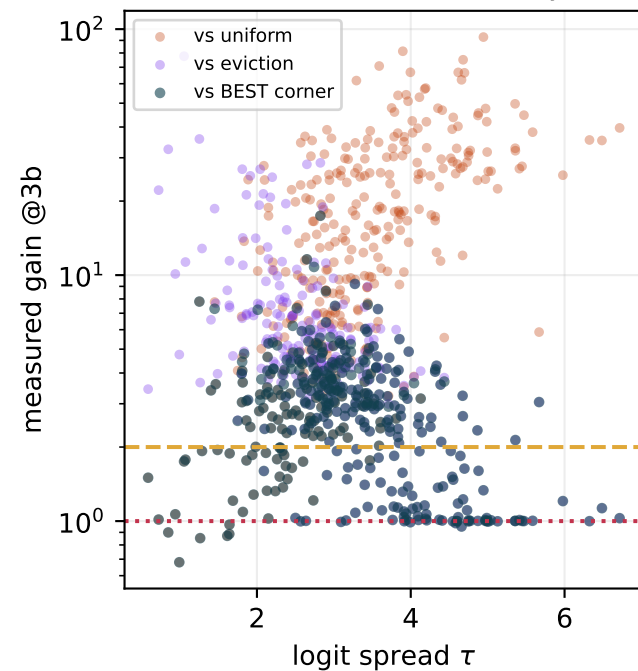
HEADS IN BAND: 69%



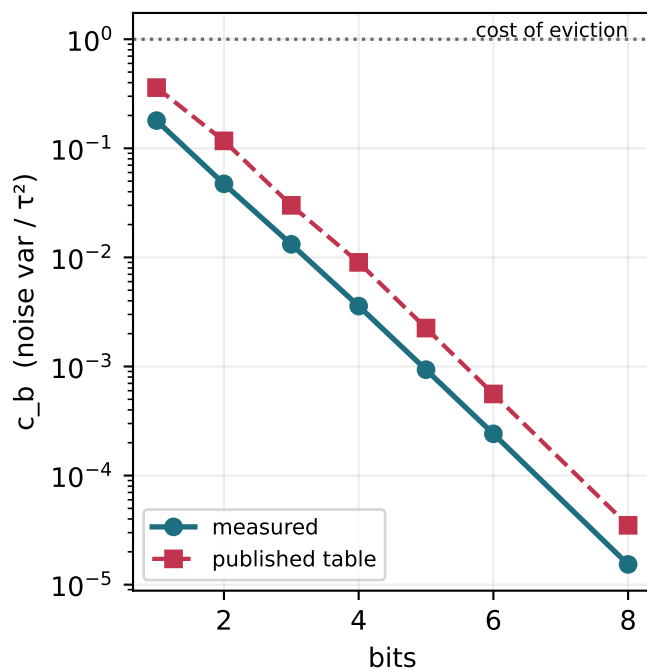
Which heads the router sends to SIEVE



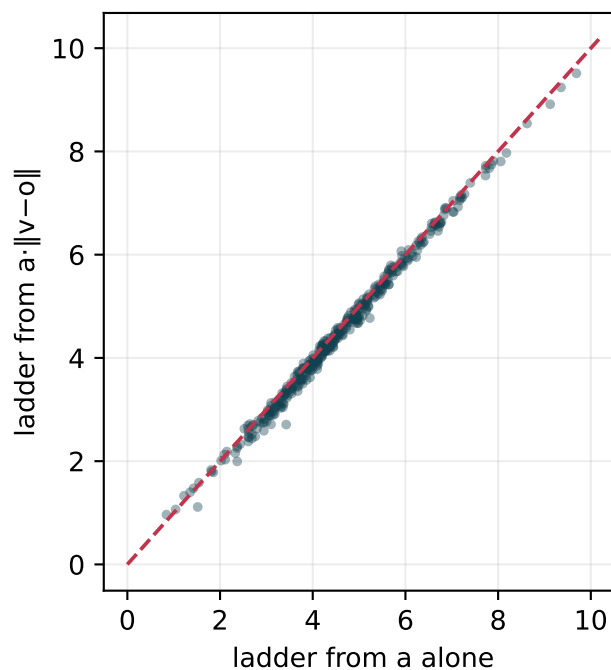
Where does the interior actually win?



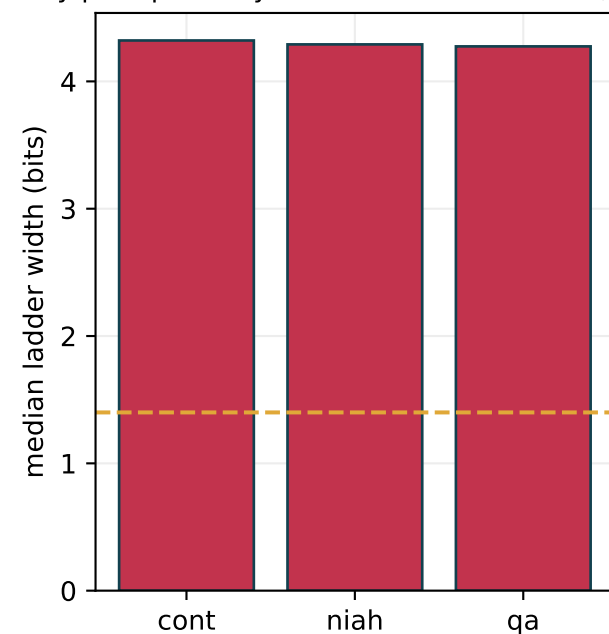
Measured vs assumed noise



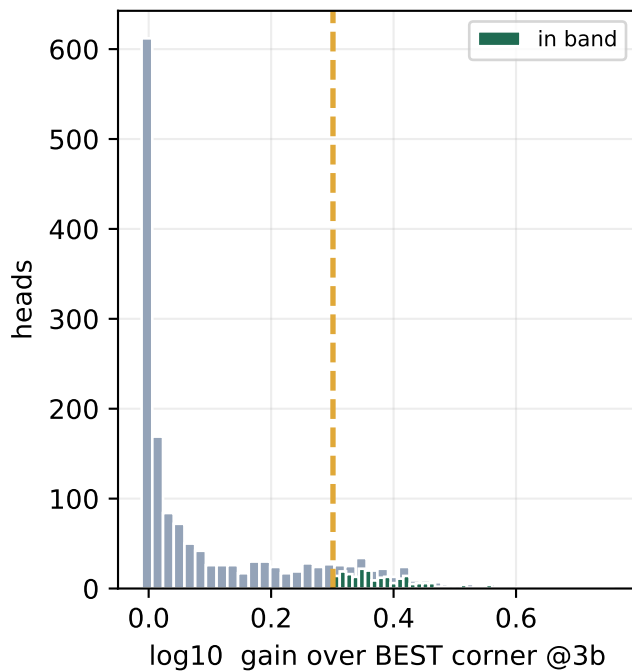
Does the value term matter?



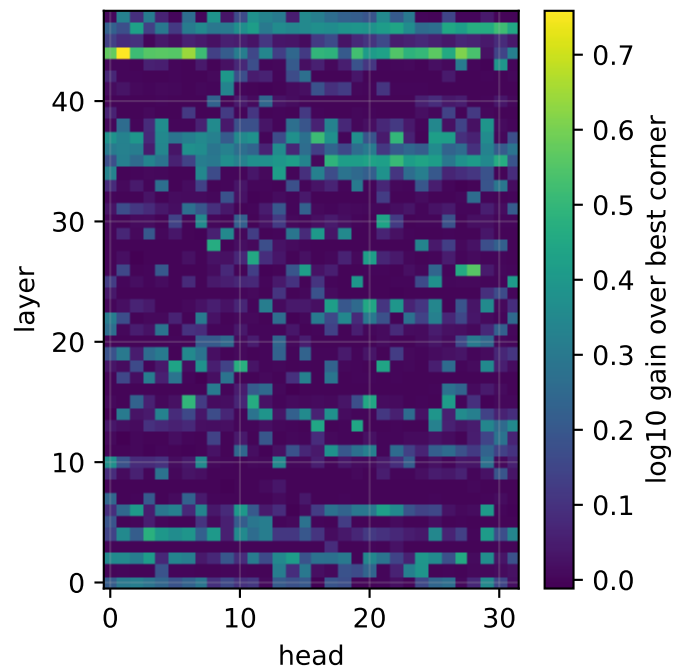
By prompt family — niah+cont +0.028 b, FA



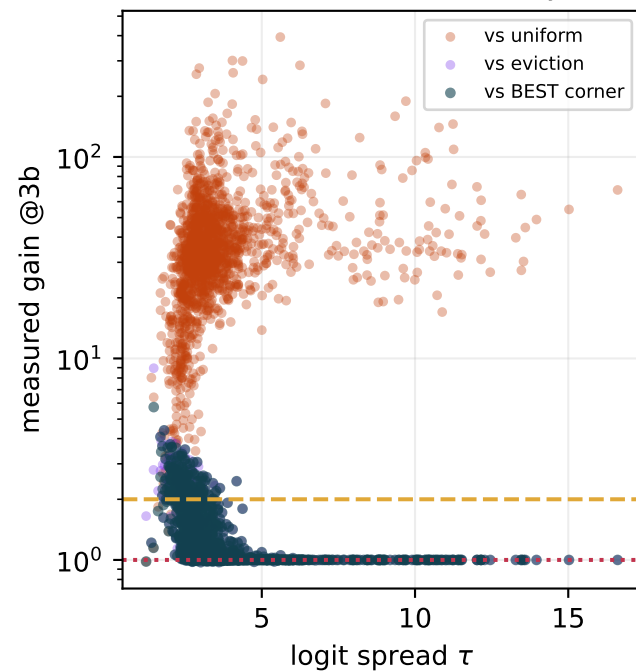
HEADS IN BAND: 14%



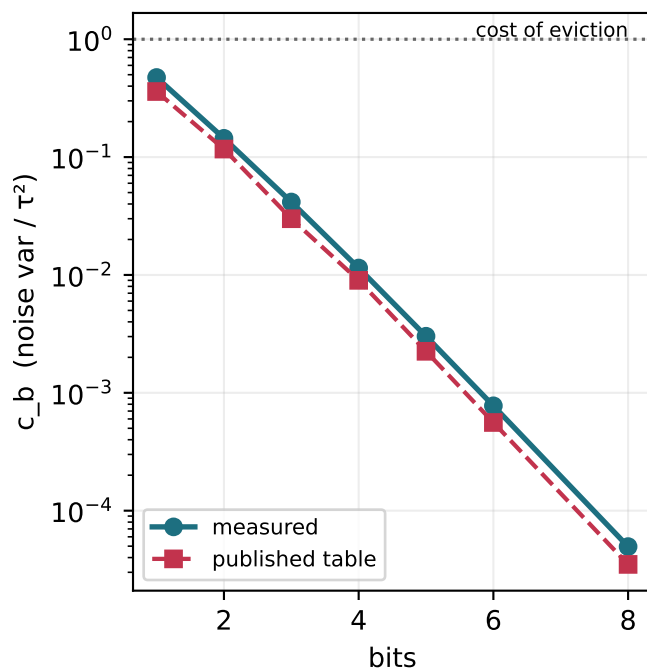
Which heads the router sends to SIEVE



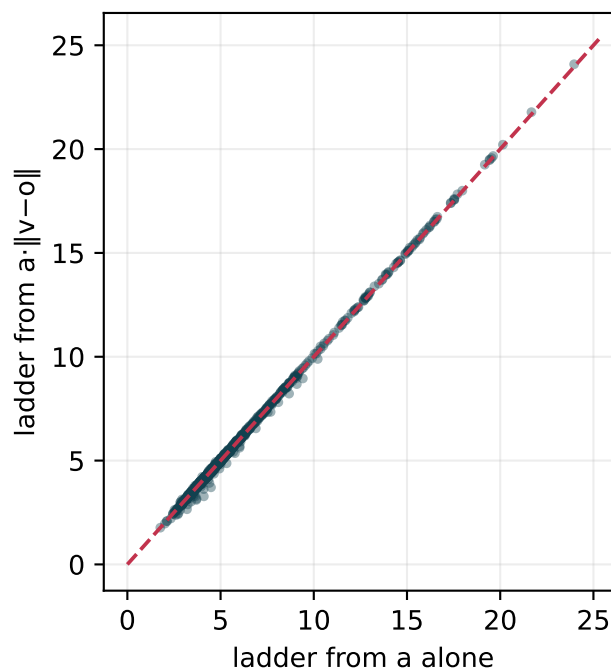
Where does the interior actually win?



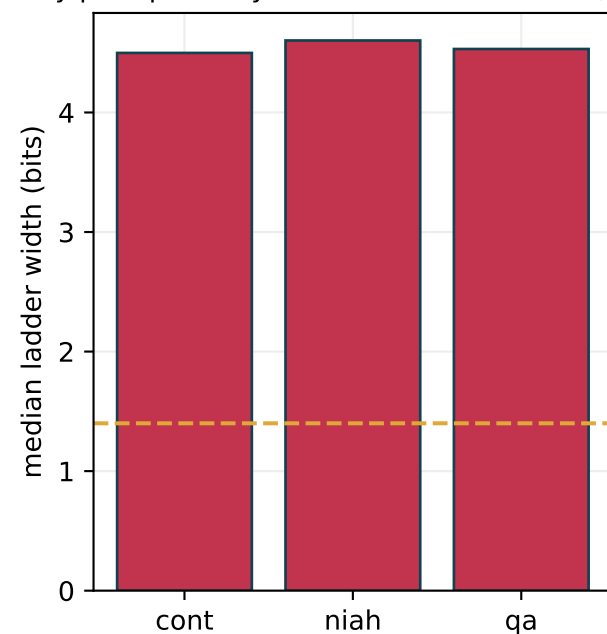
Measured vs assumed noise



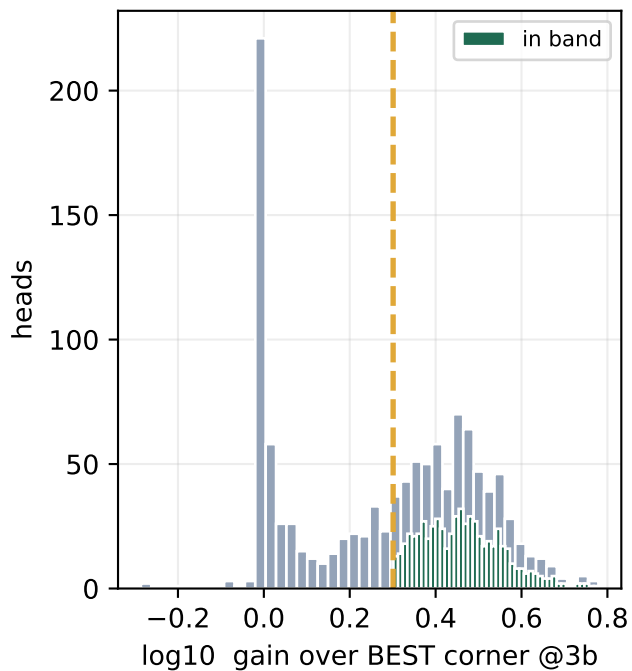
Does the value term matter?



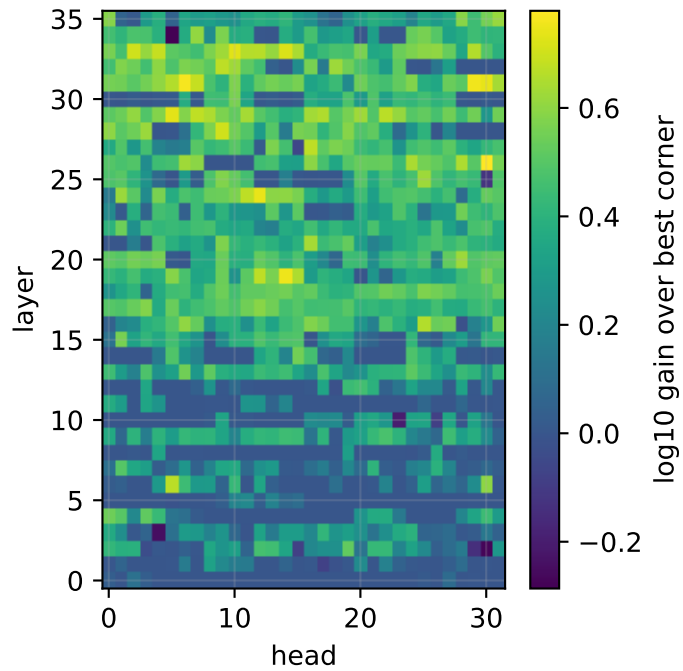
By prompt family — niah+cont +0.071 b, FA



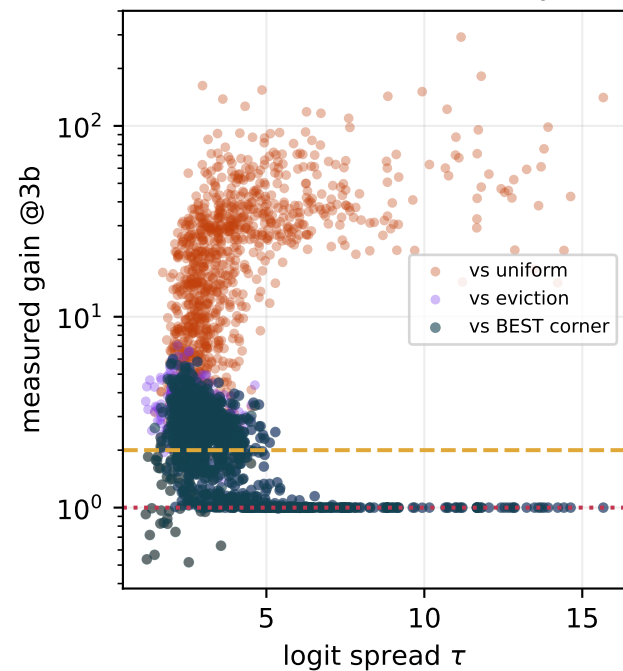
HEADS IN BAND: 54%



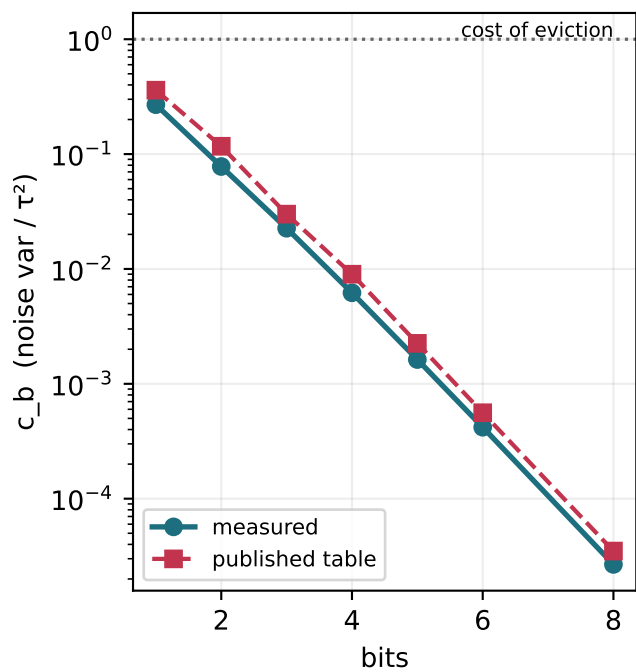
Which heads the router sends to SIEVE



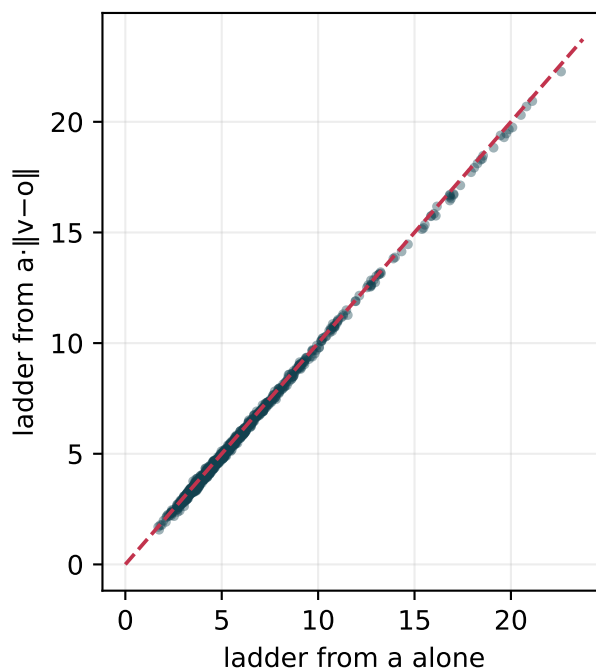
Where does the interior actually win?



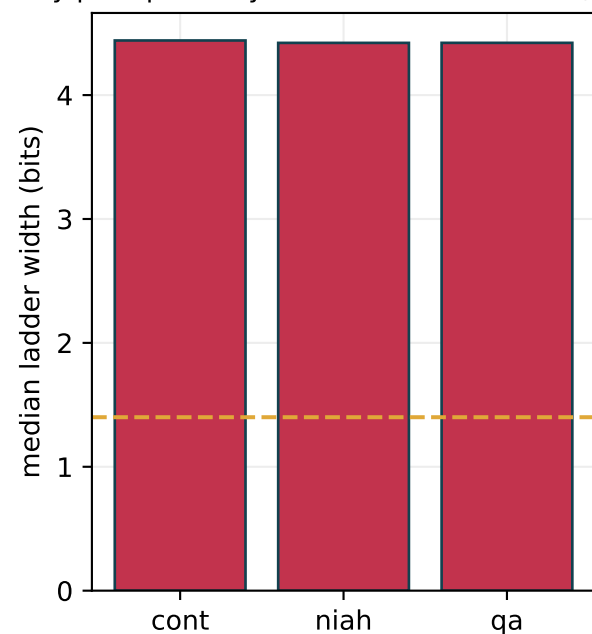
Measured vs assumed noise



Does the value term matter?



By prompt family — niah-cont -0.011 b, FA



Cross-model comparison

